MILITARY SPECIFICATION

FILLER-BLEEDER, HYDRAULIC BRAKE SYSTEM, AUTOMOTIVE

This specification is approved for use by all Department and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers a pneumatic filler-bleeder unit for servicing the hyraulic brake systems of automotive-type vehicles. The filler-bleeder unit has a diaphragm which separates the operating air pressure from the hydraulic brake fluid. The filler-bleeder has a hydraulic brake fluid reservoir tank capacity of 4 gallons plus/minus 1 gallon.

2. APPLICABLE DOCUMENTS

2.1 Government docments

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals
VV-L-800 - Lubricating Oil, General Purpose Preservative (Water Displacing, Low Temperature)
PPP-B-585 - Box, Wood, Wirebound
PPP-B-601 - Boxes, Wood, Cleated Plywood
PPP-B-636 - Box, Shipping, Fiberboard
PPP-C-843 - Cushioning Material, Cellulosic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-P-40 - Packaging and Packing of Hand Tools
PPP-T-42 - Tape, Packaging - Masking, Paper
PPP-T-60 - Tape, Packaging, Waterproof

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-C-5501 - Cap and Plug, Protective, Dust and Moisture Seal
MIL-L-21260 - Lubricating Oil, Internal Combustion Engine,
MIL-B-46176 - Brake Fluid, Silicone, Automotive All Weather, Operational and Preservative

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards For Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking For Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods
MIL-STD-1189 - Standard Department of Bar Code Symbology
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless-otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF TRANSPORTATION

DOT-FMVSS 116 - Motor Vehicle Brake Fluids

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernmental documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B40.1 - Gages -Pressure and Vacuum Indicating Dial Type - Elastic Element

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J1702 - Motor Vehicle Brake Fluid - Arctic
SAE J1703 - Motor Vehicle Brake Fluid

(Application for copies should be addressed to the Society of Automotive Engineers, Ins., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernmental standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3.1 and 6.2.1).

3.2 Design. The filler-bleeder shall be new and of the manufacturer's current commercial design. The configuration shall consist of a tank with an air chamber and fluid chamber separated by a diaphragm, a brake fluid delivery hose with control valve, and a set of master brake cylinder adapters. . . The unit shall be mounted on casters and shall include means to store the delivery hose and master brake cylinder adapters. The filler-bleeder shall be designed to meet all of the requirements specified herein. Unless otherwise specified (see 6.2.1), the filler-bleeder shall be designed for use with silicon based brake fluids and preservatives conforming to the requirements of MIL-B-46176.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specifiedf herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar netals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Reclaimed materials. The Filler-bleeder may contain reclaimed materials provided such materials will not jeopardize its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores then to the same chemical composition and physical properties as the materials originally selected for use.

3.5 Construction. Construction shall be free of defects. Construction of the filler-bleeder shall be such that parts will not work loose in service. The filler-bleeder shall withstand the strains, jars, vibrations, and other conditions incidental to universal military service, shipping and storage.

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3.5.1 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or filler-bleeder. Where practicable, all such parts shall be permanently and legibly marked with applicable part numbers in accordance with MIL-STD-130.

3.5.2 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swagged, peened, staked, or otherwise permanently deformed.

3.5.3 Threads. Screw threads shall comply with FED-STD-128 and the applicable detailed standards referenced therein.

3.5.4 Welding, brazing or soldering. Welding, brazing or soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

3.6 Performance and product characteristics.

3.6.1 Function. The filler-bleeder, when properly filled with silicon based hydraulic b-id conforming to the requirements of MIL-B-46176 and correctly pressurized with air to no less than 50 pounds prt square inch (psi) shall perform the functions of filling and bleeding the hydraulic brake systems of automotive type vehicles. The filler-bleeder, all its components and accessories, shall perform all functions specified herein without evidence of permanent deformation, structural failure, leakage, deterioration of materials, corrosion, or impairment of its initial operational capability.

3.6.2 Pressure. The filler-bleeder shall withstand an internal pressure of not less than 50 psi without damage.

3.7 Detail of components.

3.7.1 Tank. The tank shall be a metal shell of bolted flange instruction with no less than 3 and no more than 5 gallon fluid capacity. The tank shall consist of a fluid chamber and an air chamber separated by a replaceable diaphragm that is impervious to degradation from silicon based hydraulic fluids. The tank shall have an air inlet valve that accepts a standard air hose fitting used for putting air in automotive tires, an air pressure safety relief valve (see 3.7.8), a fluid pressure gage (see 3.7.7), a fluid delivery control valve (see 3.7.9), a fluid filling port with pressure tight plug (see 3.7.4), a bleeder valve (see 3.7.5), and a rolling carriage (see 3.7.2). The filler-bleeder tank shall incorporate means to store the fluid delivery hose and master cylinder adapters (see 3.7.10). The tank shall sustain 50 psi air pressure without visible deformation or more than .5 psi leakage in a 1 hour period. The filler-bleeder shall deliver not less than 95 percent of its full fluid capacity before requiring refilling. The filler-bleeder and diaphragm shall be designed to prevent damage to the diaphragm when the air chamber is charged to bypass pressure with no fluid in the fluid chamber. When air pressure is relieved from the air chamber, the diaphragm shall not block the air inlet valve or safety relief valve.

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3.7.2 Carriage. Portability shall be provided by mounting the tank assembly on at least three swivel casters. The tank shall be mounted in such a manner as to prevent tipping of the unit in normal servicing operations. The casters or wheels shall not deform or suffer damage under normal operating conditions. The caster and wheel material shall be impervious to automotive oils and greases.

3.7.3 Hose and accessory rack. The filler-bleeder shall be furnished with a means for securely storing the coiled 84 inch dispensing hose and all adapters and accessories, while the unit is being used or transported. This device may be incorporated on the tank or carriage assembly.

3.7.4 Fluid filling port. The tank shall have a fluid filling port with a hand removable cap that will prevent spillage of fluid during use and storage. The fluid filling port shall be large enough to accept the spout of a funnel to aid in filling the fluid chamber of the tank.

3.7.5 Bleeder valve. The filler-bleeder shall be equipped with a bleeder valve that enables the bleeding of any air that may be trapped in the fluid chamber.

3.7.6 Gaskets. All tank fittings and separable joints shall be equipped with gaskets made of suitable material to prevent leakage of hydraulic brake fluid and compressed air. All rubber or neoprene type components shall be designed for use with hydraulic brake fluid per SAE Standards J1702 and J17103; DOT Type 3 and 4 per DOT-FMVSS 116; and MIL-B-46176.

3.7.7 Pressure gage. The pressure gage dial shall have a diameter of not less than 1.5 inches and shall meet the requirements of ANSI B40.1, and have a Grade B accuracy. The dial shall be graduated at eaeh 5 psi and numbered at each 10 psi or more frequently. If the gage has readings above the maximum recommended operating pressure, the portion of the dial above that pressure shall be completely overprinted in red and clearly marked "DANGER". The gage shall be provided with a clear plastic window, scratch resistant, impervious to oils and hydraulic fluids, and with a screw on bezel.

3.7.8 Pressure relief valve. The filler-bleeder shall be equipped with an automatic pressure relief valve set to operate at not less than 5 psi below the manufacturer's maximum working pressure. The relief valve shall be so constructed that it cannot be set to remain closed when the pressure is greater than 50 psi. The relief valve shall be accurate within +5 psi of the setting.

3.7.9 Dispensing hose. The filler-bleeder shall be furnished with not less than 84 inches of 1/4 inch, nominal, internal diameter hose complete with fittings and a brake fluid control valve (see 3.7.1). The hose, fittings, and the brake fluid control valve shall withstand the sustained internal working pressures of not less than 50 psi. The dispensing hose shall be braided, reinforced, and shall be permanently resistant to deterioration caused by hydraulic fluids (see 3.2) and by other oils and greases associate with automotive vehicles. The dispensing hose shall be furnished with a leakproof, automatic-shutoff, quick-connect and -disconnect coupling(s) which fits its counterpart coupling on the adapters.

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3.7.10 Adapters and accessories. Unless otherwise specified (see 6.2.1), each filler-bleeder shall be furnished with the following:

a. Adapters to fit the following size automotive brake and clutch system orifices: 1/4-18 NPT, 1/2-14 NPT, 3/4-18 NS, 1-1/4-18 NEF, 1-5/8-18 NEF and a 3/4-18 NS with a minimum of two separate 45 degree connectors for vertical master cylinders.

b. Adapters to fit the following type of master cylinders:

1. Round open top
2. Open tank
3. Tandem cylinder (conventional box, vertical box and single cover)
4. Chevrolet and General Motors trucks
5. Ford Econoline trucks
6. Chrysler, Dodge Series W200 and D200 trucks
7. Special adapter to fit the master cylinder of the M151 jeep
8. Adapters to fit the master cylinder of the Commercial Utility Cargo Vehicle (CUCV) made by Chevrolet Division of General Motors
9. Adapters to fit the master cylinder of the High Mobility Multi-purpose Wheeled Vehicle (HMMWV) made by AM General Division of LTV Aerospace and Defense Company

One gasket shall be provided for each straight threaded adapter. A gasket set and a spare set containing one of each type gasket required to accommodate all adapters shall be furnished. Unless otherwise specified (see 6.2.1), each filler-bleeder shall be furnished with a minimum of two flexible bleeder tubes with a minimum length of 15 inches each. The bleeder tubes shall consist of material not subject to deterioration by contact with the hydraulic brake fluids specified in 3.2 or by contact with other oils and greases used in automotive vehicles. One tube shall be complete with connections, sizes No. 10-32 NF-2 male and 1/4-28 NF-2 male. The other tube shall provide a leakproof pressure or snap-on connector at one end.

3.8 Warning and operating instruction plate. Unless otherwise specified (see 6.2.1), a warning and operating plate shall be attached near the filler opening in a conspicuous location and shall not be affected by hydraulic brake fluid. The plate shall be permanently marked as follows:

WARNING

1. DO NOT CHARGE WITH AIR IN EXCESS OF 1/ POUNDS PER SQUARE INCH.
2. DO NOT USE SHARP TOOLS OR OTHER SHARP PROBES TO RETURN DIAPHRAGM TO ITS NORMAL DOWN POSITION.

OPERATING INSRUCTIONS

1. OPEN AIR RELEASE VALVE.
2. ADD THE SPECIFIED HYDRAULIC BRAKE FLUID.
3. CLOSE AIR RELEASE VALVE.
4. CHARGE WITH COMPRESSED AIR.
5. BLEED THE UNIT UNTIL THE UNWANTED AIR IS EVACUATED FROM THE HYDRAULIC BRAKE FLUID COMPARTMENT,

1/ Maximum working pressure shall be furnished by the manufacturer,

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3.9 Identification marking. The filler-bleeder, equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The nomenclature of the item shall be FILLER-BLEEDER, HYDRAULIC BRAKE SYSTEM, AUTOMOTIVE . Unless otherwise specified (see 6.2.1), the National Stock Number shall be included on the identification plate.

3.10 Spare parts and maintenance tools. When specified (see 6.2.1), spare parts and maintenance tools shall be furnished. Special tools shall be kept to a minimum.

3.11 Workmanship. The quality of workmanship imparted to the filler-bleeder shall equal or exceed that typically provided to commercial products by domestic producers of filler-bleeders. The filler-bleeders presented for acceptance shall have been manufactured with skill and care; shall be uniform, neat, and clean; and shall be free of irregularities and anomalies which degrade from, fit, function, performance or appearance.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective materials.

4.2 Inspection lot. All filler-bleeders offered to the Government at one time, shall be considered for purposes of inspection.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1).
b. Quality conformance inspection (see 4.3.2).

4.3.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of product examination (see 4.5) and tests (see 4.6).

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4.3.2 Quality conformance inspection. Quality conformance inspection shall be applied to production units being offered for acceptance under the contract. Quality conformance inspector shall consist of a through d as follows, and failure of any unit to pass examination, inspection or test shall be cause for rejection of the unit:

a. Product examination (4.5).
b. Air pressure retention test (4.6.2).
c. Performance test (4.6.7).
d. Packaging inspection (4.6.8).

4.4 Sampling. Each unit shall be subjected to the air pressure retention test. Sampling for production examination, performance test and packaging inspection shall be done in accordance with MIL-STD-105, Inspection Level and AQL's as specified.

		Level	AQL
a.	Product Examination (4.5)	II	1.0
b.	Air pressure Retention Test (4.6.2)	100 Percent Test	
c.	Performance Test (4.6.7)	S-4	1.0
d.	Packaging Inspection (4.6.8)	S-4	4.0

4.5 Product examination. Visually, dimensionally and manually examine each filler-bleeder to determine conformance with the requirements of 3.2 through 3.5.4, 3.7.1 through 3.7.10, and 3.8 through 3.11. Visual examination shall include verification of completeness of manufacture, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from identified defects. Manual examination shall include the operation of movable pares by 'hand at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide evidence that the machine and its components have passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances and tolerances so that no replacement part will degrade the form, fit or function of the end item (see 3.5.1) shall be cause for rejection.

4.6 Test methods. Each sample selected (see 4.4) shall be tested and any unit failing to pass the following tests, as applicable, shall be cause for rejection of the unit. Certified test reports form an independent laboratory approved by the contracting officer will be accepted as objective quality evidence that the unit has passed the required test. Tests shall be conducted as specified herein.

4.6.1 Functional and leakage tests.

4.6.1.1 Functional. The filler-bleeder shall be filled to capacity with the hydraulic brake fluid (see 3.2) and pressurized to the maximum manufacturer's working or operating pressure (see 3.6.2). Proper operation of all control valves, and the pressure gage shall be verified.

4.6.1.2 Leakage. With the filler-bleeder filled to capacity, the following sequence of operations shall be performed.

a. The unit shall be operated to discharge unwanted air in the hydraulic brake fluid compartment and dispensing hose.
b. After discharge of unwanted air, the unit shall be repressurized. The

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filler-bleeder shall be maintained in the pressurized rendition for a 15 minute period. During and after this period the filler-bleeder shall be examined for leaks around the following areas; air inlet valve, pressure gage, dispensing hose and fittings.

4.6.2 Air pressure retention test. The filler-bleeder shall be pressurized (see 3.6.2) without the use of hydraulic brake fluid and maintained in this condition for a 15 minute period without leaking as indicated by a reduction in the filler-bleeder pressure gage reading. Occurrence of leaking as evidenced by a pressure drop shall constitute failure of this test.

4.6.3 Hydrostatic pressure test. The tank, including air inlet valve, pressure gage, dispensing hose and fittings, and, when supplied, the diaphragm and automatic shut-off valve shall be subjected for a period of 15 minutes to the internal hydrostatic pressure (see 3.6.1), as applicable. During this test, the pressure relief valve shall be removed and the opening plugged, or the valve shall be made inoperative. During and after the pressure test, the tank and the above components shall be examined. Evidence of leaks, (see 3.6.1) permanent deformation, or structural failure shall constitute failure of this test.

4.6.4 Pressure gage test. To assure conformance with the requirements of ANSI B40.1, the pressure gage shall be tested at 1/4 working pressure, 1/2 working pressure, and maximum working pressure.

4.6.5 Pressure relief valve test. The filler-bleeder shall be charged with hydraulic brake fluid. The internal pressure of the filler-bleeder shall be increased until the pressure relief valve opens. Test for compliance (see 3.7.8).

4.6.6 Diaphragm test. The filler-bleeder shall be pressurized without brake fluid to withstand hydrostatic test pressure (see 3.6.2). The brake fluid flow control valve shall be opened and the filler-bleeder shall remain in this condition for 5 minutes. During and after this period, the filler-bleeder shall be examined for air leaks or diaphragm damage (see 3.6.2). Air leaks or diaphragm damage shall be cause for rejection.

4.6.7 Performance test. The filler-bleeder shall be connected for normal operation with use of a rester cylinder (or mock-up) having an opening that will accommodate one of the adapters (see 3.7.10). The master cylinder (or mock-up) shall be fitted with an outlet control valve. The filler-bleeder shall be fitted with an outlet control valve. The filler-bleeder shall be charged to capacity with hydraulic brake fluid and pressurized with air to 25 psi. With the master cylinder (or reck-up) control valve closed, the dispensing hose control valve shall be opened causing the fluid to flow under pressure into the master cylinder (or mock-up) and then draining one or two pints of brake fluid into a container. The outlet master cylinder control valve shall then be closed and the filler-bleeder repressurized at 25 psi. The equipment shall be maintained in this condition for 1 hour. During this period, the tank, dispensing hose assembly, brake fluid control valve, and adapter shall be observed for leaks. At the end of the 1 hour period, the outlet valve on the master cylinder (or mock-up) shall be opened and the hydraulic brake fluid allowed to flow into the container until fluid is no longer discharged from the filler-bleeder shall be measured and shall be at least the quantity (see 3.7.1). During the discharge, no air shall escape through the outlet master cylinder (or mock-up) control valve. Following this test, the tested unit shall be disassembled,

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thoroughly cleaned and dried reassembled and retested (see 4.6.6 and 4.6.8). Air or fluid leaks or damage to the diaphragm or other parts of the filler-bleeder shall be cause for rejection of the unit.

4.6.8 Packaging inspection. Preservation, packing, and marking of supplies and equipment for shipment and storage shall be inspected to determine compliance with Section 5.

5 PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be Level A or Level C as specified by the contract.

5.1.1 Level A.

5.1.1.1 Disassembly. When practical, for the protection of components, attachments, and accessories against damage, pilferage or to reduce cubage, items may be disassembled only to a necessary basic degree. Disassembled parts shall be preserved, anchored, braced, blocked and cushioned to prevent damage. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled item. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from assembly unless it can be reassembled readily in the field without special skills or tools.

5.1.1.2 Cleaning and drying. Thorough cleaning and drying of all bare ferrous metal surfaces is essential before the application of a preservative. Care must be taken to limit the handling of cleaned parts prior to the assembly operating. Where practicable, processors and assemblers shall wear rubber gloves while handling cleaned parts.

5.1.2 Level A.

5.1.2.1 Internal surfaces. All ferrous metal surfaces of parts which will be utilized internally in the assembly operation shall be solvent cleaned followed by fingerprint remover and wiped or drain dried. In the interim, to preclude contamination from forming on cleaned surfaces prior to the assembly operation, parts shall be coated with a light weight preservative oil, conforming to P-9 QPL VV-L-800 . To prevent dust from accumulating on cleaned parts, cover with a sheet of chemically neutral paper.

5.1.2.2 External surfaces. All accessories and attachments shall be removed prior to the external cleaning operation. All external, critical ferrous metal surfaces, fit or function, of the machine shall be solvent cleaned followed by fingerprint remover and wiped or drain dried. Noncritical ferrous metal surfaces shall be solvent cleaned and wiped or drain dried. Nonmetallic composites and painted metal surface shall be cleaned by wiping with a clean, dry cloth. Care nust be taken to prevent cleaning solvents from coming in contact with nonmetallic composites. When cleaned surfaces of the item show signs of rusting before application of the final preservative, reprocessing surface shall be confined to that area of the item. Alternatively, the solvent used for the final wiping* operation shall contain a mall mount, approximately 5 percent of volume, preservative oil conforming to P-10, type 10, grade 30, QPL MIL-L-21260. It is extremely important that elapsed time between these operations be held to an absolute minimum.

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5.1.2.3 Preservative application. Unpainted metal surfaces, threaded surfaces and surfaces exposed by disassembly shall be coated with Type P-1, preservative in accordance with MIL-P-116. Interior brake fluid surfaces shall be coated with the manufacturer's recommended hydraulic fluid by spraying of fogging through the drain and fill openings. Excess preservative shall be drained from the init.

5.1.2.4 Openings. Openings to the basic unit shall be sealed with pressure sensitive tape conforming to PPP-T-60 or by plastic plugs/caps conforming to MIL-C-5501.

5.1.2.5 Repair parts, components, and tools. Gaskets, diaphragms, and fittings shall be packaged in accordance with method IC-1 of MIL-P-116. The hose and tubing shall be coiled to a minimum safe diameter and secured. Pressure gages remaining assembled to the filler-bleeder unit shall be cushioned by wrapping with cushioning material conforming to PPP-C-843, secured with tape conforming to PPP-T-42. Gages removed during disassembly shall be wrapped with cushioning material conforming to PPP-C-843, to prevent movement, and located in a minimum sized fiberboard box conforming to PPP-B-636, class domestic, style optional. Closure of the container shall be in accordance with the appendix to specification PPP-B-636. Tools shall be packaged in accordance with PPP-P-40.

5.1.2.6 Intermediate packaging. The basic unit, with components and disassembled parts and tools, shall be located in a minimum size, close fitting fiberboard box conforming to specification PPP-B-636, class weather-resistant, style optional. Arrangement of the contents within the container and cushioning, blocking and bracing shall be in accordance with the requirements of MIL-STD-1186. Closure of the container shall be in accordance with the requirements of the appendix to specification PPP-B-636.

5.1.3 Level C. The filler-bleder shall be preserved and packaged in accordance with requirements of MIL-STD-1190.

5.2 Packing. Packing shall be level A, B or C.

5.2.1 Level A. Unless otherwise specified, a quantity of filler-bleeders packaged as specified shall be packed in shipping containers conforming to PPP-B-601, overseas type, nailed wood boxes conforming to PPP-B-585, class 3. The gross weight and size of the exterior container shall be governed by the limitations of the shipping container specification. Arrangement, cushioning, blocking and bracing of contents of the exterior shipping container shall be in accordance with the requirements of MIL-STD-1186. Closure and strapping shall be in accordance with the requirements of the appendix to the applicable box specification, except that flat steel strapping shall conform to specification QQ-S-781, type 1, class B.

5.2.2 Level B. The filler-bleeder shall be packed the sane as for Level A except that the shipping containers shall be domestic type or class 1, as applicable and class B strapping limitations need not apply.

5.2.3 Level C. Level C packing for the filler-bleeder shall be in accordance with the requirements of MIL-STD-1190.

5.3 Marking. All markings required by the contract or purchase order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Bar code markings are required and shall be applied in accordance with MIL-STD-129 and MIL-STD-1189.

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5.4 Workmanship. Workmanship shall be such that when proper procedure is followed, materials and equipment being processed will be provided the maximum protection against corrosion and deterioration consistent with the shipment hazards known or anticipated, and shall be suitable for long term storage.

6 NOTES

6.1 Intended use. This specification covers filler-bleeder intended for use in any climate normally encounter by military forces. The purpose of the filler-bleeder is to fill the master brake cylinder of automotive vehicles with the required amount of hydraulic brake fluid while the operator bleeds the wheel cylinders.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. First article, when required (see 3.1).
c. Hydraulic fluid, if different (see 3.2).
d. Master cylinder adapter fittings required (see 3.7.10 and 6.5).
e. Bleeder tubes, if different (see 3.7.10).
f. Warning and operating instruction plate, if different (see 3.8).
g. National Stock Number, when not required (see 3.9).
h. Spare parts, maintenance or special tools to be furnished (see 3.10).
i. First article, if different (see 4.3.1).
j. Selection of applicable level of preservation, packing and packing required (see 5.1 and 5.2).
k. Marking as applicable (see 5.3)

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operations and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3 Subject term (key word) listing.

Arctic brake fluid
Brake fluid
Filler-Bleeder
Silicone brake fluid

6.4 First article. When a contractor has supplied identical equipment in recent orders, the contracting officer may waive the requirement for a first article sample in the contract and arrange for an accompanying reduction in price. However, the invitation for bids include the requirement for a first article sample when the successful bidder is a contractor whose unit has not passed the first article inspection.

6.5 Master cylinder adapter data. Master cylinder adapter fittings should be identified by size and type of thread, by make and model of master cylinder, by commercial vehicle make and model, or by nomenclature of military vehicle and model number.

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AL
Air Force - 99

User activity:
Army - AV, ME
Navy - MC

Preparing activity:
Army - AL

Project No. 4910-0647

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